

The log likelihood function $\ln(L)$ of achieving this sequence of returns is:

$$\ln(L) = \sum_{i=1}^n \left(-\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\hat{\sigma}_i^2) - \frac{1}{2} \frac{y_i^2}{\hat{\sigma}_i^2} \right) \quad (6.25)$$

The parameters of the EWMA and GARCH models are found by maximizing the $\ln(L)$ where $\hat{\sigma}_i^2$ is defined from the corresponding volatility models. This maximization process can be simplified as:

$$\text{Max: } \ln(L) = \sum_{i=1}^n -\ln(\hat{\sigma}_i^2) - \frac{y_i^2}{\hat{\sigma}_i^2} \quad (6.26)$$

Many times optimization packages will only minimize an equation. In these situations, the parameters are found by minimizing the negative of the log likelihood function as follows:

$$\text{Min: } -\ln(L) = \sum_{i=1}^n \ln(\hat{\sigma}_i^2) + \frac{y_i^2}{\hat{\sigma}_i^2} \quad (6.27)$$

Estimation Results

Our parameter estimation results are shown in [Table 6.2](#). For the EWMA model, we found $\lambda = 0.88$ for both the SP500 and R2000. For the GARCH model we found $\beta = 0.84$ for the SP500 and $\beta = 0.82$ for the R2000. These findings are slightly lower than what has been previously reported for other times where λ and β are closer to 0.95 and indicate a stronger persistence. This difference is likely due to the cause of volatility persistence over our timeframe—high volatility regime caused by the debt issues followed by a lower volatility regime after the issues were resolved. The Aug. 2011 through Sep. 2011 debt issue in the US was relatively short-lived and was resolved relatively quickly.

MEASURING MODEL PERFORMANCE

We compared the HMA-VIX technique to the historical moving average (HMA), exponential weighted moving average (EWMA), and generalized autoregressive conditional heteroscedasticity (GARCH) models. We evaluated the performance of the volatility models using three different criteria: root mean square error (RMSE), root mean z-score squared error (RMZSE), and an outlier analysis. Menchero, Wang and Orr (2012) and Patton (2011) provide an in-depth discussion of alternative volatility